

INCIDENT DETECTION & AUTOMATED MITIGATION REPORT

Real-Time SSH Brute-Force Detection & Containment (Wazuh SOAR)

Document Title:	Real-Time SSH Brute-Force Detection & Containment
Target Host (Victim):	ubuntu-target (192.168.x.x)
SIEM / SOAR Manager:	wazuh-server (192.168.x.x)
Attacker Host:	Kali Linux (192.168.x.x)
Analyst & Date:	IT Manager / Cybersecurity Analyst August 18, 2026

1. Executive Summary

During continuous security monitoring with Wazuh SIEM, an automated high-severity alert was raised indicating an active SSH brute-force attack against the internal server **ubuntu-target (192.168.x.x)**. The threat actor targeted the system using Hydra from host **192.168.x.x**

Upon alert triggering, Wazuh Active Response executed an automated containment policy (**host-deny** / Active Response framework), effectively dropping and blocking the malicious IP address at the host access control layer (/etc/hosts.deny) and mitigating potential unauthorized access.

2. Incident Summary & Key Findings

Field	Details
Incident Type	SSH Password Guessing / Brute-Force Attack
Triggered Rule ID	5763 (SSHD authentication failed multiple times)
Severity Level	High / Critical (Threshold Exceeded - Level 10)
Attack Vector	Automated Credential Stuffing / Dictionary Attack via Hydra
Target User Account	analyst

Attacker IP	192.168.x.x
Target Host	ubuntu-target (192.168.x.x)
Automated Mitigation	Host-level access denial via Active Response (host-deny)
Response Time	Instantaneous (< 15 seconds)

3. Timeline of Events

- **11:16:50 UTC** — Wazuh Agent lifecycle reloaded and established connection with wazuh-server.
- **Event Window** — Attacker initiated multi-threaded SSH credential brute-force attack via Hydra (hydra -l analyst -P rockyou.txt ssh://192.168.x.x -t 4).
- **Detection Threshold** — Wazuh SIEM correlated multiple failed authentication attempts within a short interval, matching Rule ID 5763.
- **Mitigation Trigger** — Manager dispatched Active Response instruction (<location>agent</location>) to ubuntu-target.
- **Containment Execution** — The agent invoked the response script, adding 192.168.x.x to /etc/hosts.deny (or firewall drop chain). Subsequent connection attempts from the attacker host were denied.

4. Technical Analysis & Configuration Verification

A. Manager Configuration (/var/ossec/etc/ossec.conf)

The active-response policy on wazuh-server was verified as follows:

```
<command>
  <name>host-deny</name>
  <executable>host-deny</executable>
  <timeout_allowed>yes</timeout_allowed>
</command>

<active-response>
  <command>host-deny</command>
  <location>agent</location>
  <rules_id>5763</rules_id>
  <timeout>600</timeout>
</active-response>
```

B. Agent Configuration & Verification (ubuntu-target)

1. **Active Response Status:** Confirmed <disabled>no</disabled> in /var/ossec/etc/ossec.conf.

2. Execution Logs: Inspection of /var/ossec/logs/active-responses.log confirmed lifecycle communication (restart.sh agent reload) and subsequent script execution upon receiving rule triggers.

5. Containment & Verification Strategy

To verify complete containment of the threat actor:

- **Access Control List:** Confirmed entry addition in /etc/hosts.deny via command cat /etc/hosts.deny. Result: ALL: 192.168.x.x
- **Network Reachability Test (Attacker Host):** Tested connectivity from Kali Linux via ping -c 3 192.168.x.x and ssh analyst@192.168.x.x Result: Connection refused / 100% packet loss.

6. Recommendations & Post-Incident Actions

- 1. Enforce SSH Key Authentication:** Disable password-based SSH authentication (PasswordAuthentication no in /etc/ssh/sshd_config) across production nodes.
- 2. Adjust Ban Timeout Period:** Increase the active response timeout duration from 600 seconds (10 minutes) to 86400 seconds (24 hours) for repeated offenders.
- 3. Port Obfuscation:** Reconfigure the SSH listening port from standard port 22 to a non-standard high port to reduce automated script exposure.
- 4. Account Audit:** Verify that no credentials for analyst were compromised prior to the execution of the active response block.